

Course Code: CS2227 Name of the Programme: B.Tech (H) Name of the Course: AI and ML Semester: III		
Course credit - 03	No. of hours per week (L + T + P)	Total no. of Teaching hours
	2-0-2	60 Hours
Course Objectives:		
a) Fundamental principles and applications of machine learning algorithms		
b) Implementation of preprocessing techniques to clean/transform raw datasets		
c) Evaluate regression and classification model's performance on real-time datasets.		
d) Implementation of unsupervised learning algorithms for pattern discovery and analysis in datasets.		
e) Build Multilayer Perceptron to perform classification		

Course outline (Syllabus of the course) Syllabus:

Module - 1	No. of hours
	06
Foundations of Artificial Intelligence and Machine Learning Overview of Artificial Intelligence, History, Can Machines Think?, Turing Test, State Space Search, and Heuristic Search. History of machine learning, Definition and types of machine learning: supervised, unsupervised, semi-supervised, and reinforcement learning algorithms. Applications of machine learning: natural language processing, sentiment analysis, language translation, and computer vision—object recognition, object detection, and object generation.	
Module - 2	No. of hours
	06
Supervised Learning Models: Regression Linear, lasso, ridge, and polynomial regression, loss functions, MSE, MAE, RMSE, gradient descent, learning Rate, and Challenges: overfitting/underfitting and regularization.	

Module - 3	No. of hours
	06
Supervised Learning Models: Classification and Evaluation Metrics K-Nearest Neighbours (KNN): Basics, distance metrics, choice of k, SVM, Decision Trees: Splitting criteria (Gini/entropy), tree depth, Random Forest, and Ensemble Modelling. Model Evaluation: Metrics: Accuracy, precision, recall, F1 score, AUC-ROC, confusion matrix. Cross-Validation: Purpose, k-fold validation (applied to KNN/decision trees), and Interpreting Model Evaluation Metrics and Overview of Hyperparameter tuning: Grid search.	

Module - 4	No. of hours
	06
Unsupervised Learning K-means clustering, Dimensionality reduction: Principal Component Analysis (PCA), and Association Rule Mining: Apriori Algorithm	

Module - 5	No. of hours
	06
Perceptron and Feed-Forward neural networks Perceptron: logistic regression, activation function, and various types of activation functions, binary classification, error backpropagation algorithm, feed-forward neural network, Difference between feed-forward neural network and multilayer perceptron, universal approximation theorem.	

Course outcomes: Students will be able to
1. Understand various types of machine learning algorithms and the role of data preprocessing in machine learning.
2. Apply data preprocessing techniques.
3. Evaluate the regression and classification models' performance on real-time datasets.

4. Apply unsupervised learning algorithms for pattern discovery and analysis in datasets.

5. Build a Multilayer Perceptron to perform classification.

Text Books

1	James, Gareth, et al. An introduction to statistical learning. Vol. 112. New York: Springer, 2013. ISBN-10 : 1461471370
2	Müller, A. C., & Guido, S. (2016). Introduction to machine learning with Python: A Guide for data scientists. " O'Reilly Media, Inc." ISBN-13 : 978-9352134571
3	Mitchell, Tom M. Machine Learning. McGraw-Hill, 1997. ISBN-13 : 978-1259096952

Reference books

1	Aurélien, Géron. "Hands-on machine learning with scikit-learn & tensorflow." Geron Aurelien 134 (2017): 145-150. ISBN-13 : 978-9355421982
2	"Pattern Recognition and Machine Learning" by Christopher M. Bishop 2009. ISBN 13 : 978-1493938438.

Lab Programs /Tutorials

1	Perform various data preprocessing techniques like handling missing data and feature scaling.
2	Perform various data preprocessing techniques like handling outliers and encoding categorical variables.
3	Build a linear regression model to analyze the relationship between dependent and independent variables.
4	Implement K-NN to perform classification.
5	Implement SVM to perform classification.
6	Implement a Decision Tree and a Random Forest to perform classification.
7	Implement cross-fold validation to assess the performance of classification models.

8	Develop grid search and random search hyperparameter tuning methods.
9	Implement K-Means clustering algorithms to group the samples.
10	Build a multilayer perceptron to perform classification.

Lesson Plan

Name of the Programme:	B.Tech (H)	
Title of the Course:	AI and ML	
Course Code	CS2227	
Semester:	III	
Name(s) of the Course Instructor(s)	Prof. Ashwini Mathur	
Course credit	No. of hours per week (L + T + P)	Total no. of Teaching hours
3	2 + 0 + 2	30L + 0T + 30P = 60H

Session	Module No.	Topic	Pedagogy / Activities	Pre-reading / Reference
Session 1	M O D U L E I	Overview of Artificial Intelligence, History, Can Machines Think?, Turing Test	PPT and Discussion	Russell & Norvig Ch1
Session 2		State Space Search and Heuristic Search. History of machine learning.	PPT and Discussion	Russell & Norvig Ch3,4
Session 3		Lab Setup and Environment.	Hands-on Lab	Anaconda/Jupyter/Colab Docs
Session 4		Lab: Python Basics and ML Packages	Hands-on Lab	Python/NumPy/Pandas/Matplotlib Docs
Session 5		Definition and types of machine learning: supervised, unsupervised, semi-supervised. Reinforcement learning algorithms.	PPT and Discussion	Géron Ch1; Russell & Norvig Ch21
Session 6		Applications of machine learning: natural language processing, sentiment analysis, language translation, and computer vision—object recognition, object detection, and object generation.	PPT and Discussion	Géron Ch14,15

Session 7		Lab: Perform various data preprocessing techniques like handling missing data and feature scaling.	Hands-on Lab	Géron Ch2; Sklearn Preprocessing
Session 8		Lab: Continuous Assessment	Submission + VIVA	Review M1-7
Session 9		Practice Questions: Doubt Clarification for Module 01.	PPT and Discussion	M1 Review
Session 10	M O D U L E II	Introduction to Linear Regression Problem and Lasso Regression.	PPT and Discussion	Géron Ch2,4; Hastie et al. Ch3
Session 11		Lab: Project Phase 01	Presentation + Viva	Project Prep
Session 12		Lab: Project Phase 01	Presentation + Viva	Project Prep
Session 13		Ridge	PPT and Discussion	Géron Ch4; Hastie et al. Ch3
Session 14		Polynomial regression.	PPT and Discussion	Géron Ch4
Session 15		Lab: Perform various data preprocessing techniques like handling outliers and encoding categorical variables.	Hands-on Lab	Géron Ch2; Sklearn Preprocessing
Session 16		Lab: Continuous Assessment	Submission + VIVA	Review M10-15
Session 17		Loss functions, MSE, MAE, RMSE,.	PPT and Discussion	Géron Ch4; Hastie et al. Ch2
Session 18		Gradient descent and learning Rate	PPT and Discussion	Géron Ch4; Goodfellow et al. Ch4
Session 19		Gradient descent and learning Rate	PPT and Discussion	Review S18
Session 20		Lab: Build a linear regression model to analyze the relationship between dependent and independent variables.	Hands-on Lab	Géron Ch2; Sklearn LinearReg
Session 21		Lab: Continuous Assessment	Submission + VIVA	Review M17-20
Session 22		Challenges: overfitting/underfitting and regularization.	PPT and Discussion	Géron Ch4; Hastie et al. Ch7
Session 23	M	K-Nearest Neighbours (KNN): Basics, distance metrics, choice of k,	PPT and Discussion	Géron Ch3; Hastie et al. Ch2
Session 24		SVM, Decision Trees: Splitting criteria (Gini/entropy) and Tree depth	PPT and Discussion	Géron Ch5,6; Hastie et al. Ch4,9

Session 25	O D U L E III	Lab: Implement K-NN to perform classification.	Hands-on Lab	Géron Ch3; Sklearn KNN
Session 26		Lab: Continuous Assessment	Submission + VIVA	Review M23-25
Session 27		Random Forest	PPT and Discussion	Géron Ch7; Hastie et al. Ch15
Session 28		Ensemble Modelling (Along with Demonstration)	PPT and Discussion	Géron Ch7; Hastie et al. Ch10
Session 29		Lab Hour: Model Evaluation: Metrics: Accuracy, precision, recall, F1 score	Hands-on Lab	Géron Ch3; Sklearn Metrics
Session 30		Lab Hour: AUC-ROC and confusion matrix.	Submission + VIVA	Géron Ch3; Sklearn Metrics
Session 31		Cross-Validation: Purpose, k-fold validation (applied to KNN/decision trees).	PPT and Discussion	Géron Ch2; Hastie et al. Ch7
Session 32		Interpreting Model Evaluation Metrics.	PPT and Discussion	Review S29,30
Session 33		Lab: Project Phase 02	Presentation + Viva	Project Prep
Session 34		Lab: Project Phase 02	Presentation + Viva	Project Prep
Session 35		Overview of Hyperparameter tuning: Grid search.	PPT and Discussion	Géron Ch2; Sklearn GridSearchCV
Session 36		DIY Activity : Hyper Parameter Tuning	PPT and Discussion	Practical Tuning
Session 37		Lab: Implement a Decision Tree and a Random Forest to perform classification.	Hands-on Lab	Géron Ch6,7; Sklearn DT/RF
Session 38		Lab: Continuous Assessment	Submission + VIVA	Review M35-37
Session 39	M O D U	K-means clustering,	PPT and Discussion	Géron Ch9; Hastie et al. Ch14
Session 40		Dimensionality reduction: Principal Component Analysis (PCA)	PPT and Discussion	Géron Ch8; Hastie et al. Ch14
Session 41		Lab: Implement cross-fold validation to assess the performance of classification models	Hands-on Lab	Géron Ch2; Sklearn CrossVal
Session 42		Lab: Continuous Assessment	Submission + VIVA	Review M39-41

Session 43	L E IV	Association Rule Mining: Apriori Algorithm	PPT and Discussion	Tan et al. Ch6
Session 45		Practices - Doubt Clarification	PPT and Discussion	M4 Review
Session 46		Lab: Develop grid search and random search hyperparameter tuning methods.	Hands-on Lab	Géron Ch2; Sklearn Grid/RandomSearch
Session 47		Lab: Continuous Assessment	Submission + VIVA	Review M43-46
Session 48		Practices - Doubt Clarification	PPT and Discussion	M4 Review
Session 49	M O D U L E V	Perceptron and Feed-Forward neural networks. Perceptron: logistic regression.	PPT and Discussion	Géron Ch10; Goodfellow et al. Ch6
Session 50		Activation function and various types of activation functions.	PPT and Discussion	Géron Ch10; Goodfellow et al. Ch6
Session 51		Lab: Implement K-Means clustering algorithms to group the samples.	Hands-on Lab	Géron Ch9; Sklearn KMeans
		Lab: Continuous Assessment	Submission + VIVA	Review M49-51
Session 52		Binary classification and error backpropagation algorithm	PPT and Discussion	Géron Ch10; Goodfellow et al. Ch6
Session 53		Lab: Build a multilayer perceptron to perform classification.	Hands-on Lab	Géron Ch10; Keras/TF Docs
Session 54		Lab: Continuous Assessment	Submission + VIVA	Review M52-53
Session 55		Feed-forward neural network	PPT and Discussion	Goodfellow et al. Ch6; Géron Ch10
Session 56		Difference between feed-forward neural network and multilayer perceptron	PPT and Discussion	Conceptual; Goodfellow et al. Ch6
Session 57		Universal approximation theorem	PPT and Discussion	Goodfellow et al. Ch6; Wiki UAT
Session 58-60		Extra Doubt Clarification Sessions.	Discussion	Overall Review

Assessment and Evaluation

Continuous Internal Evaluation (CIE) - 70 Marks	
Components of CIE	Weightage of each component (Marks)
CIE 1 - Quiz + Theory (Pen & Paper)	20
CIE 2 - Theory (Pen & Paper)	25
CIE 3 - Lab Continuous Internal Evaluation - Mini Project _ Final Phase	10 15

Semester End Exams (SEE) - 30 Marks	
Mode of Exam	Pen and Paper
Weightage of the exam	30

Course Outcomes- Programme Outcomes Matrix

PO		PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO													
CO1	Understand various types of machine learning algorithms and the role of data preprocessing in machine learning.	3	2										

CO2	Apply data preprocessing techniques.	3	3	2									
CO3	Evaluate the regression and classification models' performance on real-time datasets.	3		3	2								
CO4	Apply unsupervised learning algorithms for pattern discovery and analysis in datasets.	3		2	3								
CO5	Build a Multilayer Perceptron to perform classification.	3	2	3		3							

Rubrics for evaluation of CIE (separate for each CIE component)

Total Marks		20
CIE Components		Rubrics for Assessment
CIE-1	Theory	Test will be conducted for 20 marks and will be evaluated according to the Scheme of Evaluation prepared by the course faculty.

Total Marks		25
CIE Components		Rubrics for Assessment
CIE-2	Theory	Test will be conducted for 25 marks and will be evaluated according to the Scheme of Evaluation prepared by the course faculty.

Total Marks		25
CIE Components		Rubrics for Assessment
CIE-3	Mini Project	Mini Project will be evaluated for 15 marks based on assessment criteria prepared by the course faculty. (<i>Refer Table 1</i>)
	Lab CIE	Lab CIE will be evaluated for 10 marks based on assessment criteria prepared by the course faculty. (<i>Refer Table 2</i>)

Table 1:

S. No	Assessment Criteria	Excellent (5 M)	Good (4 M)	Satisfactory (2-3 M)	Not Satisfactory (1 M)
1	Problem-Solving (5 Marks)	Uses advanced problem-solving skills to create innovative and effective solutions.	Uses good problem-solving skills to create effective solutions.	Uses basic problem-solving skills with partially effective solutions.	Uses poor problem-solving skills with ineffective or incorrect solutions.
2	Project demonstrations and viva voce (5 Marks)	Present the whole project in an organized manner and competently address the and handle the questions from audiences. Show the demonstration without struggling.	Present the whole project in an organized manner and competently address the and handle the questions from audiences. Show the demonstration with struggling.	Present the whole project in an organized manner and competently address the and handle the questions from audiences.	Presentation is not clear and lacked in answering the questions
3	Report (5 marks)	All the required information is included, the report is organized according to the templates and submitted with the deadline.	Most of the required information is included, the report is organized according to the templates and submitted with the deadline.	Necessary information is included with some irrelevant information. Report partially follows the given template and submission exceeds the given deadline.	The report is not organized and important information is missing. submission exceeded the given deadline.

Table 2:

S. No	Assessment Criteria	Excellent	Good	Satisfactory	Need improvement
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1	Understanding the concept (3 Marks)	Demonstrates complete understanding of concepts with clear explanations (3)	Shows good understanding with minor gaps (2)	Basic understanding with significant gaps (1)	Poor or incorrect understanding (0)
2	Implementation and Execution (3 Marks)	Code is correct, efficient, and well-structured; executes without errors (3)	Code executes correctly with minor errors or inefficiencies (2)	Code partially correct; major errors but shows effort (1)	Code incomplete or does not execute (0)
3	Viva (2 Marks)	viva answers are clear and correct (2)	viva answers mostly correct (1.5)	viva answers partially correct (1)	viva answers incorrect (0)
4	Record submission (2 Marks)	Record is complete, neat, well-presented (2)	Record is complete with minor issues (1.5)	Record incomplete or untidy (1)	Record missing or poor (0)

Signature of the Course Faculty